Operational Analytics Portfolio

Public-safe anonymized BI case studies based on previous professional work

Luis Vergara | BI Developer | Operational Analytics | Data Analyst

Public portfolio confidentiality note: This portfolio contains anonymized dashboard reconstructions based on previous professional BI work. Branding, values, names and sensitive details have been modified for confidentiality. The examples preserve the business logic and analytical approach without disclosing proprietary information.

PORTFOLIO OVERVIEW

Focus	Operational analytics, business intelligence, data pipelines, KPI reporting and decision support.
Company context	Engineering consultancy environment with multiple business units, operational teams and recurring management reporting needs.
Primary users	Team leaders, operational managers, general managers and directors/owners.
Usage cadence	Daily, weekly and monthly management routines supported through Power BI, email subscriptions and PDF versions.
Data sources	WorkflowMax timesheets and jobs, Zoho CRM sales pipeline and activities, Xero invoices and purchase orders, Excel targets and manual business inputs.
Technical role	BI owner responsible for end-to-end extraction, transformation, reporting, KPI logic, refresh routines and data quality checks.

SELECTED CASE STUDIES

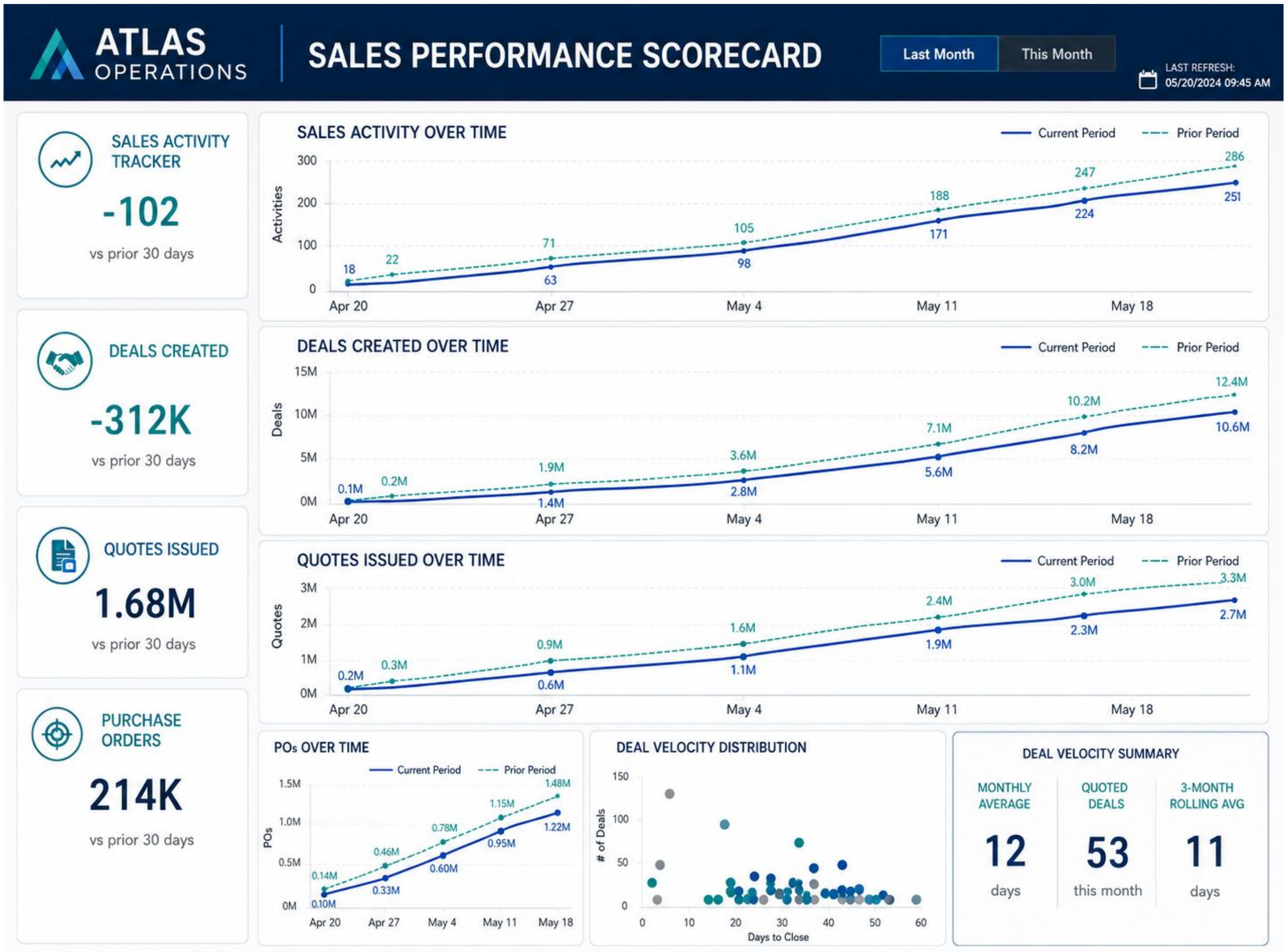
- Sales Performance Scorecard - commercial KPI visibility, sales activity tracking and forecasting.
- Business Health Check - productive capacity, utilisation and team-level business performance monitoring.
- Daily Operations Dashboard - timesheet accountability, automated reminders and revenue leakage risk analysis.

Note for recruiters and hiring managers: These case studies are designed to explain the business problems, analytical approach and operational impact behind the work. They are not intended to disclose client, employer or employee-level information.

Case Study 1 - Sales Performance Scorecard

Commercial analytics, sales activity tracking and revenue visibility

This dashboard was designed to help leadership understand whether sales activities were translating into future work and revenue. Before the reporting system, sales activity was not consistently tracked and much of the knowledge lived in informal processes or incomplete spreadsheets.



Anonymized dashboard reconstruction. Branding, values and sensitive details modified for confidentiality.

BUSINESS PROBLEM

- Sales activity, deal creation, quotes, purchase orders and conversion metrics lacked consistent visibility.
- Leadership wanted to connect controllable sales inputs, such as calls, meetings and client visits, with commercial outcomes.
- The business needed to track growth targets while avoiding overreaction to a single weak month.
- Quote speed mattered: internal analysis showed that faster quote turnaround was associated with stronger acceptance outcomes.

MY ROLE

- Designed and maintained the reporting logic with sales and leadership teams.
- Built the end-to-end pipeline using API extracts, R/Python transformations and Power BI reporting.
- Created KPI logic for sales activities, deals created, quotes issued, purchase orders, win ratio, quote velocity and deal value.
- Built data governance checks in Zoho CRM to flag missing or inconsistent critical fields for weekly review.

TECHNICAL APPROACH

- Data sources: Zoho CRM and Xero, extracted via API.
- Transformations: R, Python, Power Query and DAX.
- Modelling: structured outputs designed for Power BI consumption using star-schema-style reporting tables.

- Refresh: underlying data refreshed regularly, with Power BI scheduled refreshes throughout the day.

BUSINESS IMPACT

- Sales meetings became more focused on process improvement and action points instead of manually reviewing spreadsheets.
- Improved accountability around sales activity and visibility of commercial performance by business unit.
- Supported forecasting by connecting sales activity, quote volume, win ratio and purchase orders.
- Supported commercial reporting routines during a multi-year period of significant revenue growth.

SKILLS DEMONSTRATED

Power BI, Zoho CRM, Xero, API extraction, R, Python, DAX, sales analytics, KPI design, forecasting, stakeholder management

Case Study 2 - Business Health Check

Productive capacity, utilisation and operational health monitoring

This dashboard gave leadership a structured view of operational delivery. Since the business sold professional service hours, management needed to understand whether each team had enough work, whether capacity was being used effectively and whether billable performance aligned with targets.

Anonymized dashboard reconstruction. Branding, values and sensitive details modified for confidentiality.

BUSINESS PROBLEM

- Leadership needed a reliable way to compare available capacity against billable delivery by business unit.
- Billable and non-billable hours were difficult to interpret and sometimes did not align with expected operational performance.
- Managers needed to identify teams that were under-utilised, overworked or in need of sales support.
- The business needed a single health view combining sales context, delivery capacity and team performance.

MY ROLE

- Worked with finance and operational leaders to define realistic capacity targets.
- Designed the utilisation logic, colour-coded thresholds and team-level performance views.
- Combined timesheet data, capacity targets and team structures into a reporting model that could be reviewed by leadership.
- Created supporting checks to improve confidence in the underlying data used for management review.

TECHNICAL APPROACH

- Data sources: WorkflowMax timesheets and Excel-based capacity targets.
- Additional extraction: Selenium automation used where required reporting fields were not available through the standard API.
- Transformations: R and Python used to extract, structure and prepare tabulated files for Power BI.
- Reporting: Power BI and DAX used for utilisation, variance, team aggregation and red/yellow/green threshold logic.

BUSINESS IMPACT

- Provided management with a recurring view of operational health by team and business unit.

- Helped identify teams below target, teams under pressure and areas needing more sales or resourcing support.
- Improved visibility into billable vs non-billable work and supported financial performance conversations.
- Allowed leadership to connect sales pipeline visibility with delivery capacity and staffing decisions.

SKILLS DEMONSTRATED

Power BI, WorkflowMax, capacity modelling, utilisation analysis, Excel integration, R, Python, DAX, operational reporting, stakeholder collaboration

Case Study 3 - Daily Operations Dashboard

Timesheet accountability, automated reminders and revenue leakage risk analysis

This dashboard was built to make timesheet completion, billable hours and operational variance visible every day. The business had issues with late or incomplete timesheets, which made it harder to recover billable work accurately once jobs had progressed or been invoiced.

	DAILY						WEEKLY					
	TARGET	BILLED	NON-BILLABLE	TOTAL HOURS	VARIANCE	MISSING TIMESHEET	TARGET	BILLED	NON-BILLABLE	TOTAL HOURS	VARIANCE	MISSING TIMESHEET
DELIVERY & INTEGRATION	11.6	15.3	64.1	79.4	3.7	4x	34.2	28.6	209.7	238.3	-5.9	4xM 4xT
TECHNICAL SERVICES	45.2	40.6	7.4	48.0	-4.6	—	135.2	119.3	29.2	148.5	-16.7	—
PROJECT ENGINEERING	165.2	163.8	28.5	192.3	-2.9	1x	492.7	489.0	93.7	582.7	-3.7	1xT
OPERATIONS SUPPORT	88.2	78.6	9.8	88.4	-0.2	1x	232.6	218.2	31.1	249.3	-13.3	—
CLIENT OPERATIONS	58.0	64.1	25.1	89.2	6.1	1x	189.4	190.6	56.0	246.6	1.2	2xM 1xT
BUSINESS SUPPORT	—	—	—	—	—	—	—	—	—	—	—	—
OVERVIEW	368.2	362.4	134.9	497.3	-5.8	7x	1,084.1	1,045.7	419.7	1,464.8	-38.4	6xM 6xT

Anonymized dashboard reconstruction. Branding, values and sensitive details modified for confidentiality.

BUSINESS PROBLEM

- Some team members submitted timesheets late, sometimes only at the end of the week or month.
- Late entries increased the risk of missing billable hours, incorrect non-billable classification and invoice timing issues.
- Managers did not have a clean daily view of missing timesheets, billable hours, non-billable hours, target variance or team-level performance.
- The business needed accountability without relying on manual chasing or micromanagement.

MY ROLE

- Built the daily reporting system to show targets, billed hours, non-billable hours, total hours, variance and missing timesheets by team.
- Created automated reminder emails for missing timesheets using Outlook API / Microsoft Graph workflows.
- Designed scripts that generated daily status files and logs before team meetings.
- Built DAX measures and cleaning logic to support variance tracking, missing timesheet status and team performance review.

TECHNICAL APPROACH

- Data sources: WorkflowMax timesheets, Excel targets and supporting operational data.
- Extraction and automation: API extraction, R, Python, Selenium, batch files and Windows Scheduler.
- Refresh: regular data refresh with an early-morning automation run before daily team meetings.
- Reporting: Power BI dashboards distributed through screens, email subscriptions and PDF versions.

BUSINESS IMPACT

- Made timesheet compliance part of daily operations and gave managers a clear action list before team meetings.

- Improved accountability by making missing timesheets and variance visible at team level.
- Supported analysis of significant multi-year revenue leakage risk linked to delayed or missing billable time.
- Shifted meetings from searching for missing information to discussing clear operational actions.

SKILLS DEMONSTRATED

Power BI, WorkflowMax, R, Python, Microsoft Graph, Outlook API, Selenium, scheduled automation, DAX, operational analytics, data quality monitoring

HOW TO PRESENT THIS PORTFOLIO

- Position these as anonymized reconstructions of previous professional work, not fictional tutorial projects.
- Focus conversations on the business problem, data sources, KPI logic and operational impact rather than visual design alone.
- Use these case studies to demonstrate ownership of end-to-end BI systems, stakeholder communication, automation and operational analytics.
- Avoid claiming enterprise cloud engineering experience. The strength of the portfolio is practical business-facing analytics built from real operational data.